# Ellipse: Questions and Full Solutions

## Question 01

**English**

Let $S$ be the focus of the ellipse $\dfrac{x^2}{25}+\dfrac{y^2}{9}=1$ that lies on the positive $x$-axis, and let $D$ be the directrix of the ellipse corresponding to $S$. For a chord $PQ$ of the ellipse passing through $S$, let $\Gamma$ be the circle having $PQ$ as a diameter, let $r$ be the radius of $\Gamma$ and let $d$ be the distance of the centre of $\Gamma$ from $D$. Then, for every such chord $PQ$, $\dfrac{r}{d}$ is equal to

(A) $\dfrac{3}{5}$
(B) $1$
(C) $\dfrac{4}{5}$
(D) $\dfrac{16}{25}$

Only one option is correct.

**हिन्दी**

माना कि दीर्घवृत्त $\dfrac{x^2}{25}+\dfrac{y^2}{9}=1$ की वह नाभि $S$ है जो धनात्मक $x$-अक्ष पर स्थित है, तथा $S$ के संगत नियता $D$ है। दीर्घवृत्त की $S$ से होकर जाने वाली किसी जीवा $PQ$ के लिए, माना $\Gamma$ वह वृत्त है जिसका व्यास $PQ$ है, $r$ वृत्त $\Gamma$ की त्रिज्या है तथा $d$ वृत्त $\Gamma$ के केन्द्र की $D$ से दूरी है। तब, प्रत्येक ऐसी जीवा $PQ$ के लिए, $\dfrac{r}{d}$ बराबर है

(A) $\dfrac{3}{5}$
(B) $1$
(C) $\dfrac{4}{5}$
(D) $\dfrac{16}{25}$

केवल एक विकल्प सही है।

**Solution**

For $\dfrac{x^2}{25}+\dfrac{y^2}{9}=1$: $a=5$, $b=3$, $a^2e^2=a^2-b^2=16$, so $ae=4$ and $e=\dfrac{4}{5}$. The focus on the positive $x$-axis is $S(4,0)$ and its directrix is $D:\ x=\dfrac{a}{e}=\dfrac{25}{4}$.

Let $P(x_1,y_1)$ and $Q(x_2,y_2)$ be the ends of a chord through $S$. The circle on $PQ$ as diameter is $(x-x_1)(x-x_2)+(y-y_1)(y-y_2)=0$. Completing squares, its centre is the midpoint $M\left(\dfrac{x_1+x_2}{2},\dfrac{y_1+y_2}{2}\right)$ and its radius is $r=\dfrac{PQ}{2}$.

Since $S$ lies on the segment $PQ$, $PQ=SP+SQ$. For a point $(x,y)$ of the ellipse the distance from $S$ is $e$ times its distance from $D$, that is, $SP=e\left(\dfrac{a}{e}-x_1\right)=a-ex_1$ and $SQ=a-ex_2$.

Hence $r=\dfrac{SP+SQ}{2}=a-e\,\dfrac{x_1+x_2}{2}=a-e\,x_M=e\left(\dfrac{a}{e}-x_M\right)$.

The midpoint $M$ of a chord lies inside the ellipse, so $x_M<a<\dfrac{a}{e}$ and the distance of $M$ from the line $x=\dfrac{a}{e}$ is $d=\dfrac{a}{e}-x_M$.

Therefore $\dfrac{r}{d}=e=\dfrac{4}{5}$, the same for every chord through $S$ (including the major axis: $r=5$, $d=\dfrac{25}{4}$).

In particular $r < d$, so $\Gamma$ never meets $D$.

Option (B) is the value for a parabola (the circle then touches the directrix), (A) uses $\dfrac{b}{a}$, and (D) is $e^2$.

**Correct Option: (C)**

## Question 02

**English**

Let $P$ and $Q$ be distinct points on the ellipse $\dfrac{x^2}{16}+\dfrac{y^2}{4}=1$ such that the chord $PQ$ is parallel to the line $x+2\sqrt{3}\,y=0$. Let $O$ be the origin. Then the minimum value of $\overrightarrow{OP}\cdot\overrightarrow{OQ}$, taken over all such chords $PQ$, is

(A) $-13$
(B) $-16$
(C) $-7$
(D) $-10$

Only one option is correct.

**हिन्दी**

माना कि दीर्घवृत्त $\dfrac{x^2}{16}+\dfrac{y^2}{4}=1$ पर $P$ तथा $Q$ ऐसे भिन्न बिन्दु हैं कि जीवा $PQ$ रेखा $x+2\sqrt{3}\,y=0$ के समान्तर है। माना $O$ मूलबिन्दु है। तब ऐसी सभी जीवाओं $PQ$ पर $\overrightarrow{OP}\cdot\overrightarrow{OQ}$ का न्यूनतम मान है

(A) $-13$
(B) $-16$
(C) $-7$
(D) $-10$

केवल एक विकल्प सही है।

**Solution**

Write $P(4\cos\alpha, 2\sin\alpha)$ and $Q(4\cos\beta, 2\sin\beta)$ with $\alpha \neq \beta \pmod{2\pi}$.

Slope of $PQ$: $\dfrac{2(\sin\beta-\sin\alpha)}{4(\cos\beta-\cos\alpha)}=\dfrac{2\cdot 2\cos\frac{\alpha+\beta}{2}\sin\frac{\beta-\alpha}{2}}{4\cdot(-2)\sin\frac{\alpha+\beta}{2}\sin\frac{\beta-\alpha}{2}}=-\dfrac{1}{2}\cot\dfrac{\alpha+\beta}{2}$.

The line $x+2\sqrt{3}\,y=0$ has slope $-\dfrac{1}{2\sqrt{3}}$. Equating, $\cot\dfrac{\alpha+\beta}{2}=\dfrac{1}{\sqrt{3}}$, so $\dfrac{\alpha+\beta}{2}=\dfrac{\pi}{3}+k\pi$ and $\alpha+\beta=\dfrac{2\pi}{3}+2k\pi$. In every case $\cos(\alpha+\beta)=-\dfrac{1}{2}$.

Now $\overrightarrow{OP}\cdot\overrightarrow{OQ}=16\cos\alpha\cos\beta+4\sin\alpha\sin\beta$. Using $2\cos\alpha\cos\beta=\cos(\alpha-\beta)+\cos(\alpha+\beta)$ and $2\sin\alpha\sin\beta=\cos(\alpha-\beta)-\cos(\alpha+\beta)$:

$\overrightarrow{OP}\cdot\overrightarrow{OQ}=8[\cos(\alpha-\beta)+\cos(\alpha+\beta)]+2[\cos(\alpha-\beta)-\cos(\alpha+\beta)]=10\cos(\alpha-\beta)+6\cos(\alpha+\beta)=10\cos(\alpha-\beta)-3$.

The constraint fixes only $\alpha+\beta$; $\alpha-\beta$ can take any value with $\alpha\neq\beta$. The minimum $-13$ occurs at $\alpha-\beta=\pi$, for example $\alpha=\dfrac{5\pi}{6}$, $\beta=-\dfrac{\pi}{6}$, giving $P(-2\sqrt{3},1)$, $Q(2\sqrt{3},-1)$; then $\overrightarrow{OP}\cdot\overrightarrow{OQ}=-12-1=-13$.

Option (B) $-a^2$ would need the diameter along the major axis, which is not parallel to the given line; (C) and (D) drop or mis-sign the $6\cos(\alpha+\beta)$ term.

**Correct Option: (A)**

## Question 03

**English**

Let $P$ be a point on the ellipse $4x^2+9y^2=36$ lying in the first quadrant. The tangent to the ellipse at $P$ meets the $x$-axis at $A$ and the $y$-axis at $B$. Let $O$ be the origin. Then the minimum value of $2\,OA+3\,OB$ is

(A) 12
(B) $12\sqrt{2}$
(C) 24
(D) $6\sqrt{2}$

Only one option is correct.

**हिन्दी**

माना कि दीर्घवृत्त $4x^2+9y^2=36$ पर प्रथम चतुर्थांश में स्थित एक बिन्दु $P$ है। $P$ पर दीर्घवृत्त की स्पर्श रेखा $x$-अक्ष को $A$ पर तथा $y$-अक्ष को $B$ पर मिलती है। माना $O$ मूलबिन्दु है। तब $2\,OA+3\,OB$ का न्यूनतम मान है

(A) 12
(B) $12\sqrt{2}$
(C) 24
(D) $6\sqrt{2}$

केवल एक विकल्प सही है।

**Solution**

The ellipse is $\dfrac{x^2}{9}+\dfrac{y^2}{4}=1$; take $P(3\cos\theta, 2\sin\theta)$ with $0<\theta<\dfrac{\pi}{2}$ (first quadrant).

Tangent at $P$: $\dfrac{x\cos\theta}{3}+\dfrac{y\sin\theta}{2}=1$. Putting $y=0$: $OA=\dfrac{3}{\cos\theta}$; putting $x=0$: $OB=\dfrac{2}{\sin\theta}$.

Hence $2\,OA+3\,OB=\dfrac{6}{\cos\theta}+\dfrac{6}{\sin\theta}=6(\sec\theta+\operatorname{cosec}\theta)$.

Both terms are positive, so by AM-GM $\sec\theta+\operatorname{cosec}\theta\geq 2\sqrt{\dfrac{1}{\sin\theta\cos\theta}}=2\sqrt{\dfrac{2}{\sin 2\theta}}$, with equality when $\sec\theta=\operatorname{cosec}\theta$.

Since $\sin 2\theta\leq 1$, $2\sqrt{\dfrac{2}{\sin 2\theta}}\geq 2\sqrt{2}$, with equality when $\theta=\dfrac{\pi}{4}$.

Both equalities hold together at $\theta=\dfrac{\pi}{4}$, i.e. $P\left(\dfrac{3}{\sqrt{2}},\sqrt{2}\right)$, where $OA=3\sqrt{2}$, $OB=2\sqrt{2}$ and $2\,OA+3\,OB=6\sqrt{2}+6\sqrt{2}=12\sqrt{2}$.

Option (A) comes from stopping at $2\sqrt{\sec\theta\operatorname{cosec}\theta}\geq 2$, (C) from $\sec\theta+\operatorname{cosec}\theta\geq 4$.

**Correct Option: (B)**

## Question 04

**English**

Let $A$ be the point $(2,0)$ on the ellipse $\frac{x^2}{4}+y^2=1$. A chord $AB$ of the ellipse is such that its midpoint lies on the line $y=x$ in the first quadrant. Then the length of the chord $AB$ is

(A) $\frac{2\sqrt{17}}{5}$

(B) $\frac{4\sqrt{5}}{5}$

(C) $\frac{8}{5}$

(D) $\frac{4\sqrt{17}}{5}$

Only one option is correct.

**हिन्दी**

माना कि दीर्घवृत्त $\frac{x^2}{4}+y^2=1$ पर बिन्दु $A(2,0)$ है। दीर्घवृत्त की एक जीवा $AB$ ऐसी है कि इसका मध्यबिन्दु प्रथम चतुर्थांश में रेखा $y=x$ पर स्थित है। तब जीवा $AB$ की लम्बाई है

(A) $\frac{2\sqrt{17}}{5}$

(B) $\frac{4\sqrt{5}}{5}$

(C) $\frac{8}{5}$

(D) $\frac{4\sqrt{17}}{5}$

केवल एक विकल्प सही है।

**Solution**

Let the midpoint of $AB$ be $M(h,h)$ with $h>0$ (on $y=x$ in the first quadrant).

The chord of the ellipse with midpoint $(h,k)$ is $T=S_1$: $\frac{hx}{4}+ky=\frac{h^2}{4}+k^2$. With $k=h\neq 0$, dividing by $h$: $\frac{x}{4}+y=\frac{5h}{4}$.

The chord passes through $A(2,0)$: $\frac{2}{4}=\frac{5h}{4}$, so $h=\frac{2}{5}$ and $M\left(\frac{2}{5},\frac{2}{5}\right)$; $M$ is inside the ellipse since $\frac{1}{25}+\frac{4}{25}<1$.

The other end is $B=2M-A=\left(\frac{4}{5}-2,\frac{4}{5}\right)=\left(-\frac{6}{5},\frac{4}{5}\right)$. Check: $\frac{36/25}{4}+\frac{16}{25}=\frac{9}{25}+\frac{16}{25}=1$.

$$AB=\sqrt{\left(2+\frac{6}{5}\right)^2+\left(\frac{4}{5}\right)^2}=\sqrt{\frac{256+16}{25}}=\frac{\sqrt{272}}{5}=\frac{4\sqrt{17}}{5}.$$

The condition 'first quadrant' excludes the major axis $AA'$, whose midpoint is the origin (also on $y=x$). Option (A) is $AM$ instead of $AB$.

**Correct Option: (D)**

## Question 05

**English**

Let $P$ be a point on the line $x + y = 6$ such that the chord of contact of the tangents drawn from $P$ to the ellipse $\frac{x^2}{8} + \frac{y^2}{4} = 1$ passes through the point $(2, 0)$. Then the length of this chord of contact is

(A) $\frac{8\sqrt{2}}{3}$

(B) $\frac{4\sqrt{2}}{3}$

(C) $\frac{8}{3}$

(D) $2\sqrt{2}$

Only one option is correct.

**हिन्दी**

माना कि रेखा $x + y = 6$ पर एक बिन्दु $P$ ऐसा है कि $P$ से दीर्घवृत्त $\frac{x^2}{8} + \frac{y^2}{4} = 1$ पर खींची गई स्पर्श रेखाओं की स्पर्श जीवा बिन्दु $(2, 0)$ से होकर जाती है। तब इस स्पर्श जीवा की लम्बाई है

(A) $\frac{8\sqrt{2}}{3}$

(B) $\frac{4\sqrt{2}}{3}$

(C) $\frac{8}{3}$

(D) $2\sqrt{2}$

केवल एक विकल्प सही है।

**Solution**

Let $P(h, k)$ with $h + k = 6$. The chord of contact of $P$ with respect to $\frac{x^2}{8} + \frac{y^2}{4} = 1$ is $T = 0$: $\frac{hx}{8} + \frac{ky}{4} = 1$.

It passes through $(2, 0)$: $\frac{2h}{8} = 1$, so $h = 4$ and $k = 2$. $P(4, 2)$ is outside the ellipse because $\frac{16}{8} + \frac{4}{4} - 1 = 2 > 0$, so the tangents and the chord of contact exist.

Chord of contact of $P(4, 2)$: $\frac{4x}{8} + \frac{2y}{4} = 1$, i.e. $x + y = 2$.

Substituting $y = 2 - x$ in $x^2 + 2y^2 = 8$: $x^2 + 2(4 - 4x + x^2) = 8$, i.e. $3x^2 - 8x = 0$, so $x = 0$ or $x = \frac{8}{3}$. The points of contact are $(0, 2)$ and $\left(\frac{8}{3}, -\frac{2}{3}\right)$.

Length $= \sqrt{\left(\frac{8}{3} - 0\right)^2 + \left(-\frac{2}{3} - 2\right)^2} = \sqrt{\frac{64}{9} + \frac{64}{9}} = \frac{8\sqrt{2}}{3}$.

Option (B) is half the chord; (C) is only the difference of abscissae.

**Correct Option: (A)**

## Question 06

**English**

An ellipse $E$ has its axes parallel to the coordinate axes. The end points of one of its latera recta are $(5, 4)$ and $(5, -2)$, and the centre of $E$ lies on the line $x + y = 4$. Let the directrix of $E$ nearer to the origin be the line $x = k$. Then the value of $k$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

एक दीर्घवृत्त $E$ के अक्ष निर्देशांक अक्षों के समान्तर हैं। इसके एक नाभिलम्ब के अन्त्य बिन्दु $(5, 4)$ तथा $(5, -2)$ हैं, तथा $E$ का केन्द्र रेखा $x + y = 4$ पर स्थित है। माना $E$ की मूलबिन्दु के निकटतर नियता रेखा $x = k$ है। तब $k$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

A latus rectum is perpendicular to the major axis. The given latus rectum is the vertical segment $x = 5$, $-2 \le y \le 4$, so the major axis is horizontal, and the focus on this latus rectum is its midpoint $S(5, 1)$. Its length is $6 = \dfrac{2b^2}{a}$, so $b^2 = 3a$.

The major axis is the line $y = 1$ through $S$, so the centre is on $y = 1$. It is also on $x + y = 4$, hence $C(3, 1)$ and $CS = ae = 2$, i.e. $a^2 - b^2 = a^2e^2 = 4$.

With $b^2 = 3a$: $a^2 - 3a - 4 = 0$, $(a - 4)(a + 1) = 0$, so $a = 4$ (as $a > 0$), $b^2 = 12$, $e = \dfrac{2}{4} = \dfrac{1}{2}$.

In shifted form $E : \dfrac{(x-3)^2}{16} + \dfrac{(y-1)^2}{12} = 1$, and its directrices are $x - 3 = \pm\dfrac{a}{e} = \pm 8$, i.e. $x = 11$ (corresponding to $S$) and $x = -5$.

The distances of these lines from the origin are 11 and 5, so the directrix nearer to the origin is $x = -5$ and $k = -5$.

**Final Answer: -5**

## Question 07

**English**

Consider all ellipses $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, $a>b>0$, that pass through the point $(4,3)$. For such an ellipse, let $A$ be an end of its major axis and $B$ an end of its minor axis. Among these ellipses, let $E_0$ be the one for which the length $AB$ is the least, and let $d$ be the distance between the foci of $E_0$. Then the value of $d^2$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

बिन्दु $(4,3)$ से होकर जाने वाले सभी दीर्घवृत्तों $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, $a>b>0$, पर विचार कीजिए। ऐसे किसी दीर्घवृत्त के लिए, माना $A$ उसके दीर्घ अक्ष का एक अन्त्य बिन्दु तथा $B$ उसके लघु अक्ष का एक अन्त्य बिन्दु है। इन दीर्घवृत्तों में से माना $E_0$ वह है जिसके लिए लम्बाई $AB$ न्यूनतम है, तथा $d$ दीर्घवृत्त $E_0$ की नाभियों के बीच की दूरी है। तब $d^2$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

$A(\pm a,0)$ and $B(0,\pm b)$, so $AB^2=a^2+b^2$ for every choice of the ends. Minimising $AB$ is the same as minimising $a^2+b^2$.

The ellipse passes through $(4,3)$: $\frac{16}{a^2}+\frac{9}{b^2}=1$.

Multiply $a^2+b^2$ by this expression (which equals 1): $a^2+b^2=16+9+\frac{16b^2}{a^2}+\frac{9a^2}{b^2}$.

By AM-GM, $\frac{16b^2}{a^2}+\frac{9a^2}{b^2}\geq 2\sqrt{16\cdot 9}=24$, so $a^2+b^2\geq 49$, with equality when $\frac{16b^2}{a^2}=\frac{9a^2}{b^2}$, i.e. $\frac{b^2}{a^2}=\frac{3}{4}$.

Then $\frac{9}{b^2}=\frac{12}{a^2}$ and $\frac{28}{a^2}=1$: $a^2=28$, $b^2=21$. Since $b^2<a^2$, this ellipse belongs to the family, so $E_0:\ \frac{x^2}{28}+\frac{y^2}{21}=1$ with $AB=7$.

For $E_0$: $a^2e^2=a^2-b^2=7$, the foci are $(\pm\sqrt{7},0)$ and $d=2\sqrt{7}$, so $d^2=28$ (and $e=\frac{1}{2}$).

**Final Answer: 28**

## Question 08

**English**

Let $E$ be the ellipse $9x^2 + 25y^2 - 36x + 50y - 164 = 0$. Let $S$ be the focus of $E$ nearer to the origin, and let the directrix of $E$ corresponding to $S$ be the line $x = \lambda$. Then the value of $\lambda$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $E$ दीर्घवृत्त $9x^2 + 25y^2 - 36x + 50y - 164 = 0$ है। माना $E$ की मूलबिन्दु के निकटतर नाभि $S$ है, तथा $S$ के संगत $E$ की नियता रेखा $x = \lambda$ है। तब $\lambda$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Group terms: $9(x^2-4x)+25(y^2+2y) = 164$. Completing squares: $9(x-2)^2-36+25(y+1)^2-25 = 164$, so $9(x-2)^2 + 25(y+1)^2 = 225$.

Dividing by 225: $\dfrac{(x-2)^2}{25} + \dfrac{(y+1)^2}{9} = 1$, an ellipse with centre $C(2,-1)$, major axis along $y = -1$, $a = 5$, $b = 3$.

$a^2e^2 = 25 - 9 = 16$, so $ae = 4$ and $e = \dfrac{4}{5}$; the foci are $(2 \pm 4, -1)$, i.e. $(6,-1)$ and $(-2,-1)$.

Their distances from the origin are $\sqrt{36+1} = \sqrt{37}$ and $\sqrt{4+1} = \sqrt{5}$, so $S = (-2,-1)$, the focus to the left of the centre.

The directrix corresponding to a focus lies on the same side of the centre as that focus, at distance $\dfrac{a}{e} = \dfrac{25}{4}$ from the centre: $x = 2 - \dfrac{25}{4} = -\dfrac{17}{4}$.

Hence $\lambda = -4.25$. (The other directrix, $x = 8.25$, belongs to the focus $(6,-1)$.)

**Final Answer: -4.25**

## Question 09

**English**

An ellipse has its focus at $S(1,2)$, the corresponding directrix is the line $3x+4y+9=0$, and the length of its major axis is 7.5. Let $(\alpha,\beta)$ be the vertex of the ellipse nearer to this directrix. Then the value of $\alpha+\beta$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

एक दीर्घवृत्त की नाभि $S(1,2)$ है, संगत नियता रेखा $3x+4y+9=0$ है, तथा इसके दीर्घ अक्ष की लम्बाई $7.5$ है। माना इस नियता के निकटतर दीर्घवृत्त का शीर्ष $(\alpha,\beta)$ है। तब $\alpha+\beta$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

For a focus and its directrix, $p = SZ = \dfrac{a}{e} - ae = \dfrac{a(1-e^2)}{e}$, where $Z$ is the foot of the perpendicular from $S$ to the directrix. Here $p = \dfrac{|3\cdot 1 + 4\cdot 2 + 9|}{\sqrt{9+16}} = \dfrac{20}{5} = 4$.

The major axis has length $2a = 7.5$, so $a = \dfrac{15}{4}$. Then $\dfrac{15}{4}(1-e^2) = 4e$, i.e. $15e^2 + 16e - 15 = 0$, which factorises as $(5e-3)(3e+5) = 0$; since $0 < e < 1$, $e = \dfrac{3}{5}$.

The axis of the ellipse is the line through $S$ perpendicular to the directrix, with unit direction $\pm\dfrac{(3,4)}{5}$. The value $3x+4y+9$ at $S$ is $20 > 0$, so $\dfrac{(3,4)}{5}$ points from the directrix towards $S$.

The vertex $A$ nearer to the directrix lies between $S$ and $Z$, with $SA = a - ae = a(1-e) = \dfrac{15}{4}\cdot\dfrac{2}{5} = \dfrac{3}{2}$.

So $A = S - \dfrac{3}{2}\cdot\dfrac{(3,4)}{5} = (1-0.9,\ 2-1.2) = (0.1,\ 0.8)$.

Check with the definition: distance of $A$ from the directrix is $\dfrac{0.3+3.2+9}{5} = 2.5$ and $SA = 1.5 = \dfrac{3}{5}\cdot 2.5$.

Hence $\alpha+\beta = 0.1 + 0.8 = 0.9$.

**Final Answer: 0.9**

## Question 10

**English**

Let $S$ be the origin. Let $\mathcal{P}$ be the parabola with focus $S$ and directrix $3x + 4y + 50 = 0$, and let $\mathcal{E}$ be the ellipse with focus $S$, corresponding directrix $3x + 4y + 175 = 0$ and eccentricity $\frac{1}{2}$. Let $P$ and $Q$ be the two points common to $\mathcal{P}$ and $\mathcal{E}$. Then the area of the triangle $SPQ$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $S$ मूलबिन्दु है। माना $\mathcal{P}$ वह परवलय है जिसकी नाभि $S$ तथा नियता $3x + 4y + 50 = 0$ है, तथा $\mathcal{E}$ वह दीर्घवृत्त है जिसकी नाभि $S$, संगत नियता $3x + 4y + 175 = 0$ तथा उत्केन्द्रता $\frac{1}{2}$ है। माना $\mathcal{P}$ तथा $\mathcal{E}$ के दो उभयनिष्ठ बिन्दु $P$ तथा $Q$ हैं। तब त्रिभुज $SPQ$ का क्षेत्रफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Both directrices are perpendicular to the direction $(3, 4)$, so both conics have the axis through $S$ in that direction. For a point $X(x, y)$ put $u = \dfrac{3x + 4y}{5}$; the distance of $X$ from $3x + 4y + c = 0$ is $\dfrac{|5u + c|}{5}$.

$\mathcal{P}$: $SX = |u + 10|$. If $u < -10$ this would give $SX = -u - 10 < |u| \le SX$, impossible, so on $\mathcal{P}$, $SX = u + 10$.

$\mathcal{E}$: $SX = \dfrac{1}{2}|u+35|$. If $u < -35$ then $SX = \dfrac{-u - 35}{2} < -u \le SX$, impossible, so on $\mathcal{E}$, $SX = \dfrac{u + 35}{2}$.

At a common point $u + 10 = \dfrac{u + 35}{2}$, so $u = 15$: every common point lies on the line $3x + 4y = 75$ and has $SX = 25$.

Solve $x^2 + y^2 = 625$, $y = \dfrac{75 - 3x}{4}$: $16x^2 + (75 - 3x)^2 = 10000 \Rightarrow 25x^2 - 450x - 4375 = 0 \Rightarrow x^2 - 18x - 175 = 0 \Rightarrow (x - 25)(x + 7) = 0$.

So $P(25, 0)$ and $Q(-7, 24)$. Check $Q$: $SQ = 25$; distance to $3x+4y+50 = 0$ is $\dfrac{-21 + 96 + 50}{5} = 25$; distance to $3x + 4y + 175 = 0$ is $\dfrac{250}{5} = 50 = 2\,SQ$.

Area of $\triangle SPQ$ with $S$ at the origin: $\dfrac{1}{2}|x_P y_Q - x_Q y_P| = \dfrac{1}{2}|25 \cdot 24 - (-7) \cdot 0| = 300$.

**Final Answer: 300**

## Question 11

**English**

Let $E$ be the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ and let $P_1 = \left(4, \frac{12}{5}\right)$. For each $n \geq 1$, the normal to $E$ at $P_n$ meets the $x$-axis at $G_n$, and $P_{n+1}$ is the point of $E$ in the first quadrant whose abscissa is equal to the abscissa of $G_n$. Let $x_n$ denote the abscissa of $P_n$. Then the value of $\sum_{n=1}^{\infty} x_n$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $E$ दीर्घवृत्त $\frac{x^2}{25} + \frac{y^2}{16} = 1$ है तथा $P_1 = \left(4, \frac{12}{5}\right)$ है। प्रत्येक $n \geq 1$ के लिए, $P_n$ पर $E$ का अभिलम्ब $x$-अक्ष को $G_n$ पर मिलता है, तथा $P_{n+1}$ प्रथम चतुर्थांश में $E$ का वह बिन्दु है जिसका भुज $G_n$ के भुज के बराबर है। माना $x_n$ बिन्दु $P_n$ का भुज है। तब $\sum_{n=1}^{\infty} x_n$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

$P_1$ is on $E$: $\frac{16}{25} + \frac{144/25}{16} = \frac{16}{25} + \frac{9}{25} = 1$. Write a first-quadrant point as $P_n(5\cos\theta_n, 4\sin\theta_n)$, $0 < \theta_n < \frac{\pi}{2}$.

The normal at the point with eccentric angle $\theta$ is $\frac{ax}{\cos\theta} - \frac{by}{\sin\theta} = a^2 - b^2$, here $\frac{5x}{\cos\theta_n} - \frac{4y}{\sin\theta_n} = 9$.

Putting $y = 0$: $x = \frac{9\cos\theta_n}{5} = \frac{9}{25} \cdot 5\cos\theta_n = \frac{9}{25}x_n$. So $G_n = \left(\frac{9}{25}x_n, 0\right)$; note $\frac{9}{25} = e^2$.

Since $0 < \frac{9}{25}x_n < 5$, there is exactly one point of $E$ in the first quadrant with this abscissa, so $P_{n+1}$ is well defined and $x_{n+1} = \frac{9}{25}x_n$.

Thus $x_n = 4\left(\frac{9}{25}\right)^{n-1}$, a geometric progression with ratio less than 1, and $\sum_{n=1}^{\infty} x_n = \frac{4}{1 - \frac{9}{25}} = \frac{100}{16} = \frac{25}{4} = 6.25$.

**Final Answer: 6.25**

## Question 12

**English**

Let $S$ be the focus of the ellipse $\frac{x^2}{9}+\frac{y^2}{5}=1$ that lies on the positive $x$-axis. For a point $P$ on the ellipse, let $M$ be the foot of the perpendicular from $P$ to the line $y=\frac{5}{2}$. Then the maximum value of $SP+PM$, as $P$ varies on the ellipse, is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि दीर्घवृत्त $\frac{x^2}{9}+\frac{y^2}{5}=1$ की वह नाभि $S$ है जो धनात्मक $x$-अक्ष पर स्थित है। दीर्घवृत्त पर किसी बिन्दु $P$ के लिए, माना $P$ से रेखा $y=\frac{5}{2}$ पर डाले गए लम्ब का पाद $M$ है। तब $P$ के दीर्घवृत्त पर बदलने पर $SP+PM$ का अधिकतम मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Here $a^2=9$, $b^2=5$, $a^2e^2=4$, so $ae=2$, $e=\frac{2}{3}$ and $S=(2,0)$. Write $P(3\cos\theta,\sqrt{5}\sin\theta)$, $\theta\in[0,2\pi)$.

For a point of the ellipse, $SP=e\cdot$ (distance from the directrix $x=\frac{9}{2}$) $=\frac{2}{3}\left(\frac{9}{2}-x\right)=3-\frac{2}{3}x=3-2\cos\theta$.

Every point of the ellipse has $y\le\sqrt{5}<\frac{5}{2}$, so $PM=\frac{5}{2}-y=\frac{5}{2}-\sqrt{5}\sin\theta$ (no absolute value is needed).

Hence $SP+PM=\frac{11}{2}-\left(2\cos\theta+\sqrt{5}\sin\theta\right)$.

For $A\cos\theta+B\sin\theta$ the range is $\left[-\sqrt{A^2+B^2},\sqrt{A^2+B^2}\right]$; here $\sqrt{4+5}=3$. The maximum of $SP+PM$ is $\frac{11}{2}+3=\frac{17}{2}$.

It is attained when $(\cos\theta,\sin\theta)=-\frac{1}{3}(2,\sqrt{5})$, i.e. at $P\left(-2,-\frac{5}{3}\right)$ (an end of the other latus rectum): $SP=\sqrt{16+\frac{25}{9}}=\frac{13}{3}$, $PM=\frac{5}{2}+\frac{5}{3}=\frac{25}{6}$, sum $\frac{51}{6}=8.5$.

(The minimum, $\frac{11}{2}-3=2.5$, is simply the distance of $S$ from the line, which is why the maximum is the real question.)

**Final Answer: 8.5**

## Question 13

**English**

Let $E$ be the ellipse $\frac{x^2}{16}+\frac{y^2}{12}=1$. Let $P$ be the point of $E$ with eccentric angle $\theta \in \left(\pi, \frac{3\pi}{2}\right)$ such that the tangent to $E$ at $P$ is parallel to the chord of $E$ whose midpoint is $\left(1, \frac{3}{2}\right)$. The normal to $E$ at $P$ meets the $x$-axis at $(\alpha, 0)$. Then the value of $\alpha$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $E$ दीर्घवृत्त $\frac{x^2}{16}+\frac{y^2}{12}=1$ है। माना $P$, $E$ का वह बिन्दु है जिसका उत्केन्द्र कोण $\theta \in \left(\pi, \frac{3\pi}{2}\right)$ है तथा $P$ पर $E$ की स्पर्श रेखा, $E$ की उस जीवा के समान्तर है जिसका मध्यबिन्दु $\left(1, \frac{3}{2}\right)$ है। $P$ पर $E$ का अभिलम्ब $x$-अक्ष को $(\alpha, 0)$ पर मिलता है। तब $\alpha$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

The chord of $E$ with midpoint $(h,k)$ is $T=S_1$: $\frac{hx}{16}+\frac{ky}{12}=\frac{h^2}{16}+\frac{k^2}{12}$. For $\left(1,\frac{3}{2}\right)$, which is inside $E$ since $\frac{1}{16}+\frac{9/4}{12}=\frac{1}{4}<1$: $\frac{x}{16}+\frac{y}{8}=\frac{1}{4}$, i.e. $x+2y=4$, slope $-\frac{1}{2}$.

For $P(4\cos\theta, 2\sqrt{3}\sin\theta)$ the tangent is $\frac{x\cos\theta}{4}+\frac{y\sin\theta}{2\sqrt{3}}=1$, with slope $-\frac{2\sqrt{3}\cos\theta}{4\sin\theta}=-\frac{\sqrt{3}}{2}\cot\theta$.

Parallel: $\frac{\sqrt{3}}{2}\cot\theta=\frac{1}{2}$, so $\cot\theta=\frac{1}{\sqrt{3}}$, giving $\theta=\frac{\pi}{3}$ or $\frac{4\pi}{3}$. Only $\theta=\frac{4\pi}{3}$ lies in $\left(\pi,\frac{3\pi}{2}\right)$, so $P=\left(4\cdot\left(-\frac{1}{2}\right), 2\sqrt{3}\cdot\left(-\frac{\sqrt{3}}{2}\right)\right)=(-2,-3)$.

(Indeed $P$ lies on the line through the centre and the midpoint, $y=\frac{3}{2}x$, as it must: the chords bisected on a diameter are parallel to the tangent at its end.)

Normal at $(x_1,y_1)$: $\frac{a^2x}{x_1}-\frac{b^2y}{y_1}=a^2-b^2$, here $\frac{16x}{-2}-\frac{12y}{-3}=4$, i.e. $-8x+4y=4$, or $y=2x+1$.

At $y=0$: $x=-\frac{1}{2}$. So $\alpha=-0.5$ (equivalently $\alpha=e^2x_1=\frac{1}{4}\cdot(-2)$).

**Final Answer: -0.5**

## Question 14

**English**

Let $P$ and $Q$ be the points of the ellipse $\frac{x^2}{48}+\frac{y^2}{16}=1$ with eccentric angles $\frac{\pi}{2}$ and $-\frac{\pi}{6}$, respectively. Let $R$ be a point on the ellipse. Then the maximum value of the area of the triangle $PQR$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि दीर्घवृत्त $\frac{x^2}{48}+\frac{y^2}{16}=1$ के बिन्दु $P$ तथा $Q$ के उत्केन्द्र कोण क्रमशः $\frac{\pi}{2}$ तथा $-\frac{\pi}{6}$ हैं। माना $R$ दीर्घवृत्त पर एक बिन्दु है। तब त्रिभुज $PQR$ के क्षेत्रफल का अधिकतम मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Here $a=4\sqrt{3}$, $b=4$. $P=(4\sqrt{3}\cos\frac{\pi}{2},4\sin\frac{\pi}{2})=(0,4)$ and $Q=\left(4\sqrt{3}\cdot\frac{\sqrt{3}}{2},4\cdot\left(-\frac{1}{2}\right)\right)=(6,-2)$.

The chord joining the points with eccentric angles $\alpha,\beta$ is $\frac{x}{a}\cos\frac{\alpha+\beta}{2}+\frac{y}{b}\sin\frac{\alpha+\beta}{2}=\cos\frac{\alpha-\beta}{2}$.

With $\frac{\alpha+\beta}{2}=\frac{\pi}{6}$ and $\frac{\alpha-\beta}{2}=\frac{\pi}{3}$: $\frac{x}{4\sqrt{3}}\cdot\frac{\sqrt{3}}{2}+\frac{y}{4}\cdot\frac{1}{2}=\frac{1}{2}$, i.e. $x+y=4$.

$PQ=\sqrt{36+36}=6\sqrt{2}$.

For $R(4\sqrt{3}\cos t,4\sin t)$, the distance from $x+y=4$ is $\frac{|4\sqrt{3}\cos t+4\sin t-4|}{\sqrt{2}}=\frac{|8\cos(t-\frac{\pi}{6})-4|}{\sqrt{2}}$.

Since $8\cos(t-\frac{\pi}{6})-4\in[-12,4]$, the largest distance is $\frac{12}{\sqrt{2}}=6\sqrt{2}$, at $t-\frac{\pi}{6}=\pi$, i.e. $t=\frac{7\pi}{6}$, $R(-6,-2)$ (the tangent there is parallel to $PQ$, on the far side).

Maximum area $=\frac{1}{2}\cdot PQ\cdot 6\sqrt{2}=\frac{1}{2}\cdot 6\sqrt{2}\cdot 6\sqrt{2}=36$. (The point $t=\frac{\pi}{6}$ on the near arc gives only $\frac{1}{2}\cdot 6\sqrt{2}\cdot 2\sqrt{2}=12$.)

**Final Answer: 36**

## Question 15

**English**

Let $E$ be the ellipse $\frac{x^2}{49}+\frac{y^2}{4}=1$ and let $C$ be its auxiliary circle. Then the number of points $(m, n)$, where $m$ and $n$ are integers, that lie strictly inside $C$ and strictly outside $E$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $E$ दीर्घवृत्त $\frac{x^2}{49}+\frac{y^2}{4}=1$ है तथा $C$ इसका सहायक वृत्त है। तब ऐसे बिन्दुओं $(m, n)$, जहाँ $m$ तथा $n$ पूर्णांक हैं, की संख्या जो $C$ के पूर्णतः अन्दर तथा $E$ के पूर्णतः बाहर स्थित हैं, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

The auxiliary circle of $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is $x^2+y^2=a^2$, here $C:\ x^2+y^2=49$.

A point $(m, n)$ is strictly inside $C$ iff $m^2+n^2<49$, and strictly outside $E$ iff $S_1=\frac{m^2}{49}+\frac{n^2}{4}-1>0$, i.e. $4m^2+49n^2>196$.

Inside $C$ forces $|n|\le 6$. Count by rows (the count for $-n$ equals that for $n$):

$n=0$: need $m^2<49$ and $4m^2>196$, i.e. $m^2>49$: impossible, 0 points.

$n=\pm1$: need $m^2<48$ and $4m^2>147$, i.e. $m^2>36.75$; the only candidate squares are 36 and 49, neither in range: 0 points.

$n=\pm2$: need $m^2<45$ and $4m^2>0$, i.e. $m\neq 0$, $|m|\le 6$: 12 points for each sign of $n$.

$|n|\ge 3$: $49n^2\ge 441>196$, so every such point is outside $E$; inside $C$: $n=\pm3$: $m^2<40$, $|m|\le 6$, 13 points; $n=\pm4$: $m^2<33$, $|m|\le 5$, 11; $n=\pm5$: $m^2<24$, $|m|\le 4$, 9; $n=\pm6$: $m^2<13$, $|m|\le 3$, 7.

Total $=2(0+0+12+13+11+9+7)=104$ (checked by direct enumeration).

**Final Answer: 104**

## Question 16

**English**

Let $S$ be the point $(3, 0)$ and let $L$ be the line $x = \dfrac{16}{3}$. Let $T$ be the set of all integers $\lambda$ for which the point $P(\lambda, \lambda + 1)$ satisfies $PS < \dfrac{3}{4}\, PN$, where $N$ is the foot of the perpendicular from $P$ to $L$. Then the sum of all the elements of $T$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $S$ बिन्दु $(3, 0)$ है तथा $L$ रेखा $x = \dfrac{16}{3}$ है। माना $T$ उन सभी पूर्णांकों $\lambda$ का समुच्चय है जिनके लिए बिन्दु $P(\lambda, \lambda + 1)$ प्रतिबन्ध $PS < \dfrac{3}{4}\, PN$ को सन्तुष्ट करता है, जहाँ $N$ बिन्दु $P$ से $L$ पर डाले गए लम्ब का पाद है। तब $T$ के सभी अवयवों का योग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

The locus $PS = \dfrac{3}{4} PN$ is the ellipse with focus $S(3, 0)$, directrix $x = \dfrac{16}{3}$ and $e = \dfrac{3}{4}$. For the standard ellipse centred at the origin, $ae = 3$ and $\dfrac{a}{e} = \dfrac{16}{3}$ give $a^2 = 16$, $a = 4$, and $b^2 = a^2(1 - e^2) = 16 \cdot \dfrac{7}{16} = 7$.

For any point $(x, y)$: $PS^2 - \dfrac{9}{16} PN^2 = (x - 3)^2 + y^2 - \dfrac{9}{16}\left(x - \dfrac{16}{3}\right)^2 = \dfrac{7}{16} x^2 + y^2 - 7 = 7\left(\dfrac{x^2}{16} + \dfrac{y^2}{7} - 1\right)$.

So $PS < \dfrac{3}{4} PN$ exactly when $S_1 = \dfrac{x^2}{16} + \dfrac{y^2}{7} - 1 < 0$, i.e. when $P$ lies inside the ellipse.

For $P(\lambda, \lambda + 1)$: $7\lambda^2 + 16(\lambda + 1)^2 < 112$, i.e. $23\lambda^2 + 32\lambda - 96 < 0$.

The roots are $\dfrac{-32 \pm \sqrt{1024 + 8832}}{46} = \dfrac{-32 \pm \sqrt{9856}}{46}$, about $-2.85$ and $1.46$. Check the boundary integers: $\lambda = -3$ gives $207 - 96 - 96 = 15 > 0$; $\lambda = 2$ gives $92 + 64 - 96 = 60 > 0$; $\lambda = -2$ gives $92 - 64 - 96 < 0$; $\lambda = 1$ gives $23 + 32 - 96 < 0$.

So $T = \{-2, -1, 0, 1\}$ and the sum is $-2$.

**Final Answer: -2**

## Question 17

**English**

The tangent to the curve $y = \dfrac{2}{1+x^2}$, $x \in \mathbb{R}$, at the point where $x = 1$ touches the ellipse $\dfrac{x^2}{3}+\dfrac{y^2}{b^2} = 1$, $0 < b < \sqrt{3}$, at the point $(\alpha, \beta)$. Then the value of $\beta$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

वक्र $y = \dfrac{2}{1+x^2}$, $x \in \mathbb{R}$, के उस बिन्दु पर, जहाँ $x = 1$ है, खींची गई स्पर्श रेखा दीर्घवृत्त $\dfrac{x^2}{3} + \dfrac{y^2}{b^2} = 1$, $0 < b < \sqrt{3}$, को बिन्दु $(\alpha, \beta)$ पर स्पर्श करती है। तब $\beta$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

At $x = 1$: $y = \dfrac{2}{2} = 1$. $\dfrac{dy}{dx} = -\dfrac{4x}{(1+x^2)^2}$, so the slope at $x = 1$ is $-\dfrac{4}{4} = -1$.

Tangent to the curve: $y - 1 = -(x - 1)$, i.e. $y = -x + 2$ ($m = -1$, $c = 2$).

The line $y = mx + c$ touches $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ iff $c^2 = a^2m^2 + b^2$. Here $4 = 3 \cdot 1 + b^2$, so $b^2 = 1$ (and $0 < 1 < \sqrt{3}$).

The point of contact of the tangent $y = mx+c$ is $\left(-\dfrac{a^2m}{c}, \dfrac{b^2}{c}\right) = \left(\dfrac{3}{2}, \dfrac{1}{2}\right)$. Check: it is on $x+y = 2$ and $\dfrac{9/4}{3} + \dfrac{1}{4} = \dfrac{3}{4} + \dfrac{1}{4} = 1$.

Hence $\beta = \dfrac{1}{2} = 0.5$.

**Final Answer: 0.5**

## Question 18

**English**

Let $E$ be the ellipse $\frac{x^2}{25} + \frac{y^2}{11} = 1$. For a real number $c$, let $L_c$ be the line $3x + 4y = c$. Let $c_0 < 0$ be such that there is exactly one point on $L_{c_0}$ from which the two tangents drawn to $E$ are perpendicular to each other. Then the value of $c_0$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $E$ दीर्घवृत्त $\frac{x^2}{25} + \frac{y^2}{11} = 1$ है। किसी वास्तविक संख्या $c$ के लिए, माना $L_c$ रेखा $3x + 4y = c$ है। माना $c_0 < 0$ ऐसा है कि $L_{c_0}$ पर ठीक एक ऐसा बिन्दु है जिससे $E$ पर खींची गई दोनों स्पर्श रेखाएँ परस्पर लम्बवत हैं। तब $c_0$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

The tangent of slope $m$ to $E$ is $y = mx \pm \sqrt{25m^2 + 11}$. Through $(h, k)$: $(k - mh)^2 = 25m^2 + 11$, i.e. $(h^2 - 25)m^2 - 2hkm + k^2 - 11 = 0$.

The two tangents are perpendicular iff the product of the roots is $-1$: $\frac{k^2 - 11}{h^2 - 25} = -1$, i.e. $h^2 + k^2 = 36$ (the director circle; the vertical-horizontal pairs from $(\pm 5, \pm\sqrt{11})$ also lie on it).

So the required points of $L_c$ are its intersections with $x^2 + y^2 = 36$. There is exactly one when $L_c$ is tangent to this circle: distance of the origin from $3x + 4y = c$ equals 6, $\frac{|c|}{5} = 6$, $c = \pm 30$.

With $c_0 < 0$: $c_0 = -30$; the point is $-6 \cdot \frac{(3, 4)}{5} = \left(-\frac{18}{5}, -\frac{24}{5}\right)$.

**Final Answer: -30**

## Question 19

**English**

Let $E$ be the ellipse $\frac{x^2}{19}+\frac{y^2}{6}=1$. Let $P$ be the point in the first quadrant on the line $4x=3y$ from which the two tangents drawn to $E$ are perpendicular to each other. If $m_1$ and $m_2$ are the slopes of these two tangents, then the value of $m_1+m_2$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $E$ दीर्घवृत्त $\frac{x^2}{19}+\frac{y^2}{6}=1$ है। माना रेखा $4x=3y$ पर प्रथम चतुर्थांश में $P$ वह बिन्दु है जिससे $E$ पर खींची गई दोनों स्पर्श रेखाएँ परस्पर लम्बवत हैं। यदि इन दोनों स्पर्श रेखाओं की प्रवणताएँ $m_1$ तथा $m_2$ हैं, तो $m_1+m_2$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

From $(h,k)$ the slopes of the tangents to $E$ satisfy $(k-mh)^2=19m^2+6$, i.e. $(h^2-19)m^2-2hkm+(k^2-6)=0$. The tangents are perpendicular iff $\frac{k^2-6}{h^2-19}=-1$, i.e. $h^2+k^2=25$ (director circle).

With $k=\frac{4}{3}h$: $h^2+\frac{16}{9}h^2=25$, $h^2=9$, and in the first quadrant $P(3,4)$. $P$ is outside $E$: $\frac{9}{19}+\frac{16}{6}>1$.

For $P(3,4)$: $(9-19)m^2-24m+(16-6)=0$, i.e. $-10m^2-24m+10=0$, or $5m^2+12m-5=0$.

The discriminant $144+100>0$ gives two real slopes with $m_1m_2=-1$, as required, and $m_1+m_2=-\frac{12}{5}=-2.4$.

**Final Answer: -2.4**

## Question 20

**English**

From the point $P(2, 4)$, two tangents are drawn to the ellipse $\frac{x^2}{21} + \frac{y^2}{15} = 1$. These tangents meet the $x$-axis at the points $A$ and $B$. Then the area of the triangle $PAB$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

बिन्दु $P(2, 4)$ से दीर्घवृत्त $\frac{x^2}{21} + \frac{y^2}{15} = 1$ पर दो स्पर्श रेखाएँ खींची जाती हैं। ये स्पर्श रेखाएँ $x$-अक्ष को बिन्दुओं $A$ तथा $B$ पर मिलती हैं। तब त्रिभुज $PAB$ का क्षेत्रफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

$P(2, 4)$ is outside the ellipse since $\frac{4}{21} + \frac{16}{15} - 1 = \frac{20 + 112 - 105}{105} > 0$, so two tangents exist. Neither is vertical, because $x = 2$ is not a tangent ($2^2 \neq 21$).

A line of slope $m$ through $P$ is $y = mx + (4 - 2m)$; it touches the ellipse iff $(4 - 2m)^2 = 21m^2 + 15$, i.e. $16 - 16m + 4m^2 = 21m^2 + 15$, so $17m^2 + 16m - 1 = 0$.

$(17m - 1)(m + 1) = 0$: $m_1 = \frac{1}{17}$, $m_2 = -1$. Check: for $m = -1$, $c = 6$ and $c^2 = 36 = 21 + 15$.

Tangent with slope $\frac{1}{17}$: $y = \frac{x + 66}{17}$, meeting the $x$-axis at $A(-66, 0)$. Tangent with slope $-1$: $y = 6 - x$, meeting the $x$-axis at $B(6, 0)$.

$AB = 72$ and the height of $P$ above the $x$-axis is 4, so the area of $\triangle PAB$ is $\frac{1}{2} \cdot 72 \cdot 4 = 144$.

**Final Answer: 144**

## Question 21

**English**

Let $M$ be the midpoint of a chord of the ellipse $\frac{x^2}{4}+\frac{y^2}{3}=1$ whose slope is $\frac{1}{2}$. If $M$ lies on the line $x-y=1$, then the abscissa of $M$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि दीर्घवृत्त $\frac{x^2}{4}+\frac{y^2}{3}=1$ की एक जीवा, जिसकी प्रवणता $\frac{1}{2}$ है, का मध्यबिन्दु $M$ है। यदि $M$ रेखा $x-y=1$ पर स्थित है, तो $M$ का भुज _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

The chord of the ellipse with midpoint $M(h,k)$ is $T=S_1$: $\frac{hx}{4}+\frac{ky}{3}=\frac{h^2}{4}+\frac{k^2}{3}$, whose slope is $-\frac{3h}{4k}$ (for $k\neq 0$).

Slope $\frac{1}{2}$: $-\frac{3h}{4k}=\frac{1}{2}$, so $k=-\frac{3h}{2}$. ($k=0$ would give a vertical chord, so $k\neq 0$.)

$M$ on $x-y=1$: $h+\frac{3h}{2}=1$, so $h=\frac{2}{5}$ and $k=-\frac{3}{5}$.

$M$ is inside the ellipse: $\frac{4/25}{4}+\frac{9/25}{3}=\frac{4}{25}<1$, so the chord exists; its ends are $x=\frac{2\pm 3\sqrt{7}}{5}$ on $y=\frac{x}{2}-\frac{4}{5}$, with midpoint abscissa $\frac{2}{5}$.

Hence the abscissa of $M$ is 0.4.

**Final Answer: 0.4**

## Question 22

**English**

Let $E$ be the ellipse $\frac{x^2}{18}+\frac{y^2}{8}=1$ and let $L$ be the polar of the point $Q(3,1)$ with respect to $E$. Let $(\alpha,\beta)$ be the point of $L$ whose chord of contact with respect to $E$ has $Q$ as its midpoint. Then the value of $\alpha+\beta$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $E$ दीर्घवृत्त $\frac{x^2}{18}+\frac{y^2}{8}=1$ है तथा $E$ के सापेक्ष बिन्दु $Q(3,1)$ की ध्रुवीय रेखा $L$ है। माना $L$ का वह बिन्दु $(\alpha,\beta)$ है जिसकी $E$ के सापेक्ष स्पर्श जीवा का मध्यबिन्दु $Q$ है। तब $\alpha+\beta$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

$Q(3,1)$ is inside $E$ since $\frac{9}{18}+\frac{1}{8}=\frac{5}{8}<1$. Its polar is $T(Q)=0$: $\frac{3x}{18}+\frac{1\cdot y}{8}=1$, i.e. $L:\ 4x+3y=24$, a line lying outside $E$.

For a point $R(\alpha,\beta)$ outside $E$, the chord of contact is its polar $\frac{\alpha x}{18}+\frac{\beta y}{8}=1$.

The chord of $E$ with midpoint $Q$ is $T=S_1$: $\frac{3x}{18}+\frac{y}{8}=\frac{9}{18}+\frac{1}{8}=\frac{5}{8}$, i.e. $\frac{8}{5}\cdot\frac{3x}{18}+\frac{8}{5}\cdot\frac{y}{8}=1$.

The chord of contact of $R$ is this chord iff $\frac{\alpha}{18}=\frac{8}{5}\cdot\frac{3}{18}$ and $\frac{\beta}{8}=\frac{8}{5}\cdot\frac{1}{8}$, i.e. $R=\frac{8}{5}Q=\left(\frac{24}{5},\frac{8}{5}\right)$.

$R$ lies on $L$: $4\cdot\frac{24}{5}+3\cdot\frac{8}{5}=\frac{120}{5}=24$. This is no accident: the chord of contact of $R$ passes through $Q$, so by the pole-polar reciprocity $R$ lies on the polar of $Q$. $R$ is outside $E$ ($\frac{576/25}{18}+\frac{64/25}{8}=\frac{8}{5}>1$), so its chord of contact exists.

Hence $\alpha+\beta=\frac{24}{5}+\frac{8}{5}=6.4$.

**Final Answer: 6.4**

## Question 23

**English**

The pair of tangents drawn from the point $P(3, 6)$ to the ellipse $\frac{x^2}{7} + \frac{y^2}{2} = 1$ meets the $y$-axis at the points $A$ and $B$. Then the area of the triangle $PAB$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

बिन्दु $P(3, 6)$ से दीर्घवृत्त $\frac{x^2}{7} + \frac{y^2}{2} = 1$ पर खींचा गया स्पर्श रेखा युग्म $y$-अक्ष को बिन्दुओं $A$ तथा $B$ पर मिलता है। तब त्रिभुज $PAB$ का क्षेत्रफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

$P(3, 6)$ is outside the ellipse: $S_1 = \frac{9}{7} + \frac{36}{2} - 1 = \frac{128}{7} > 0$.

The combined equation of the pair of tangents from $P$ is $SS_1 = T^2$, where $S = \frac{x^2}{7} + \frac{y^2}{2} - 1$ and $T = \frac{3x}{7} + \frac{6y}{2} - 1 = \frac{3x}{7} + 3y - 1$.

Points of this pair on the $y$-axis: put $x = 0$: $\frac{128}{7}\left(\frac{y^2}{2} - 1\right) = (3y-1)^2$. Multiply by 7: $64y^2 - 128 = 63y^2 - 42y + 7$, so $y^2 + 42y - 135 = 0$.

$(y - 3)(y + 45) = 0$: the tangents meet the $y$-axis at $A(0, 3)$ and $B(0, -45)$ (one on each tangent, since the pair meets $x = 0$ in two distinct points and $P$ is not on $x = 0$).

Check with slopes: the tangents are $y = x + 3$ (slope 1, $c^2 = 9 = 7 + 2$) and $y = 17x - 45$ (slope 17, $c^2 = 2025 = 7 \cdot 289 + 2$).

$AB = 48$ and the distance of $P$ from the $y$-axis is 3, so the area of $\triangle PAB$ is $\frac{1}{2} \cdot 48 \cdot 3 = 72$.

**Final Answer: 72**

## Question 24

**English**

Let $\theta$ be the acute angle between the two tangents drawn from the point $P(4,4)$ to the ellipse $\frac{x^2}{10}+\frac{y^2}{6}=1$. Then the value of $\tan\theta$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि बिन्दु $P(4,4)$ से दीर्घवृत्त $\frac{x^2}{10}+\frac{y^2}{6}=1$ पर खींची गई दोनों स्पर्श रेखाओं के बीच का न्यून कोण $\theta$ है। तब $\tan\theta$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

$S_1=\frac{16}{10}+\frac{16}{6}-1=\frac{24+40-15}{15}=\frac{49}{15}>0$, so $P$ is outside and two tangents exist.

Slopes of the tangents from $P$: $y=mx+(4-4m)$ touches the ellipse iff $(4-4m)^2=10m^2+6$, i.e. $6m^2-32m+10=0$, or $3m^2-16m+5=0$; so $m_1+m_2=\frac{16}{3}$, $m_1m_2=\frac{5}{3}$ (no tangent is vertical, as $16\neq 10$).

Equivalently, the pair of tangents is $SS_1=T^2$ with $T=\frac{4x}{10}+\frac{4y}{6}-1$; its second-degree part is $Ax^2+2Hxy+By^2$ with $A=\frac{49}{150}-\frac{16}{100}=\frac{1}{6}$, $B=\frac{49}{90}-\frac{16}{36}=\frac{1}{10}$, $H=-\frac{4}{15}$. The lines through $P$ are parallel to $Ax^2+2Hxy+By^2=0$, so their slopes satisfy $Bm^2+2Hm+A=0$, i.e. $3m^2-16m+5=0$, agreeing with the direct computation.

$$\tan\theta=\left|\frac{m_1-m_2}{1+m_1m_2}\right|=\frac{\sqrt{(m_1+m_2)^2-4m_1m_2}}{1+m_1m_2}=\frac{\sqrt{\frac{256}{9}-\frac{20}{3}}}{\frac{8}{3}}=\frac{\frac{14}{3}}{\frac{8}{3}}=\frac{7}{4}.$$

Since $1+m_1m_2>0$, the angle given by this value is acute, so $\tan\theta=1.75$. (Same value from $\frac{2\sqrt{H^2-AB}}{|A+B|}=\frac{2\cdot\frac{7}{30}}{\frac{4}{15}}$.)

**Final Answer: 1.75**

## Question 25

**English**

The ellipse $E: \dfrac{x^2}{18}+\dfrac{y^2}{9}=1$ is a reflecting curve. A ray of light starting from the point $S(3,0)$ travels along the line joining $S$ to the point $B(0,3)$ and is reflected by $E$ at $B$. The reflected ray meets $E$ again at the point $Q$. The tangent to $E$ at $Q$ meets the $x$-axis at $(\alpha,0)$. Then the value of $\alpha$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

दीर्घवृत्त $E: \dfrac{x^2}{18}+\dfrac{y^2}{9}=1$ एक परावर्तक वक्र है। बिन्दु $S(3,0)$ से चलने वाली एक प्रकाश किरण $S$ को बिन्दु $B(0,3)$ से मिलाने वाली रेखा के अनुदिश चलती है तथा $B$ पर $E$ द्वारा परावर्तित होती है। परावर्तित किरण $E$ को पुनः बिन्दु $Q$ पर मिलती है। $Q$ पर $E$ की स्पर्श रेखा $x$-अक्ष को $(\alpha,0)$ पर मिलती है। तब $\alpha$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

For $E$: $a^2=18$, $b^2=9$, $a^2e^2=9$, so the foci are $S(3,0)$ and $S'(-3,0)$; $B(0,3)$ is an end of the minor axis, on $E$.

Reflection property: the normal at a point of the ellipse bisects the angle between the focal radii, so a ray coming from one focus is reflected towards the other focus. Hence the reflected ray at $B$ travels along $BS'$.

Check at $B$: the tangent at $B$ is $y=3$ and the normal is vertical; the incoming direction $(-3,3)$ reflects to $(-3,-3)$, which points from $B$ to $S'(-3,0)$.

Line $BS'$: $y=x+3$. With $x^2+2y^2=18$: $x^2+2(x^2+6x+9)=18$, i.e. $3x^2+12x=0$, so $x=0$ (the point $B$) or $x=-4$. Hence $Q(-4,-1)$; check $\dfrac{16}{18}+\dfrac{1}{9}=1$.

Tangent at $Q(x_1,y_1)$: $\dfrac{xx_1}{18}+\dfrac{yy_1}{9}=1$, i.e. $-\dfrac{4x}{18}-\dfrac{y}{9}=1$, or $2x+y=-9$.

At $y=0$: $x=-\dfrac{9}{2}$, so $\alpha=-4.5$ (equivalently $\alpha=\dfrac{a^2}{x_1}=\dfrac{18}{-4}$).

**Final Answer: -4.5**

## Question 26

**English**

Let $P$ be the point of the ellipse $\frac{x^2}{20} + \frac{y^2}{5} = 1$ with eccentric angle $\frac{\pi}{4}$, and let $S$ and $S'$ be the foci of the ellipse. Let $U$ and $V$ be the feet of the perpendiculars from $S$ and $S'$, respectively, to the tangent to the ellipse at $P$. Let $O$ be the origin. Then the value of $\overrightarrow{OU} \cdot \overrightarrow{OV}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि दीर्घवृत्त $\frac{x^2}{20} + \frac{y^2}{5} = 1$ का वह बिन्दु $P$ है जिसका उत्केन्द्र कोण $\frac{\pi}{4}$ है, तथा $S$ और $S'$ दीर्घवृत्त की नाभियाँ हैं। माना $P$ पर दीर्घवृत्त की स्पर्श रेखा पर $S$ तथा $S'$ से डाले गए लम्बों के पाद क्रमशः $U$ तथा $V$ हैं। माना $O$ मूलबिन्दु है। तब $\overrightarrow{OU} \cdot \overrightarrow{OV}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Here $a^2 = 20$, $b^2 = 5$, $a^2e^2 = 15$, so $S(\sqrt{15}, 0)$, $S'(-\sqrt{15}, 0)$. $P = \left(2\sqrt{5} \cdot \frac{1}{\sqrt{2}}, \sqrt{5} \cdot \frac{1}{\sqrt{2}}\right) = (\sqrt{10}, \sqrt{\frac{5}{2}})$.

Tangent at the point with eccentric angle $\theta$: $\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1$; for $\theta = \frac{\pi}{4}$: $\frac{x}{2\sqrt{10}} + \frac{y}{\sqrt{10}} = 1$, i.e. $x + 2y = 2\sqrt{10}$.

Property: the foot of the perpendicular from a focus to any tangent lies on the auxiliary circle $x^2 + y^2 = a^2$. So $U$ and $V$ lie on $x^2 + y^2 = 20$ and on the tangent. They are distinct (the feet from $S$ and $S'$ coincide only if the tangent is perpendicular to the major axis), so they are exactly the two intersection points of the tangent with the circle.

Distance of $O$ from the tangent: $p = \frac{2\sqrt{10}}{\sqrt{1+4}} = 2\sqrt{2}$. Half the chord $UV$ is $\sqrt{a^2 - p^2} = \sqrt{20 - 8} = 2\sqrt{3}$, so $UV^2 = 48$.

$$\overrightarrow{OU} \cdot \overrightarrow{OV} = \frac{OU^2 + OV^2 - UV^2}{2} = \frac{20 + 20 - 48}{2} = -4.$$

Direct check: the foot from $S(\sqrt{15}, 0)$ on $x + 2y = 2\sqrt{10}$ is $S - \frac{\sqrt{15} - 2\sqrt{10}}{5}(1, 2)$, and $|OU|^2 = 20$; similarly for $V$, and the dot product computed from the coordinates is $-4$.

**Final Answer: -4**